

Natural Language

Natural Language

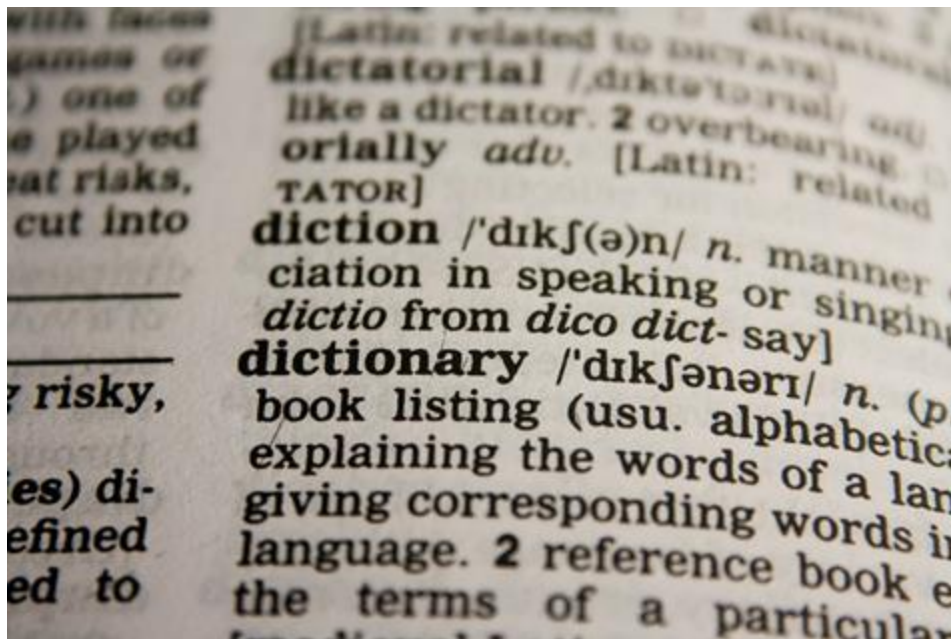
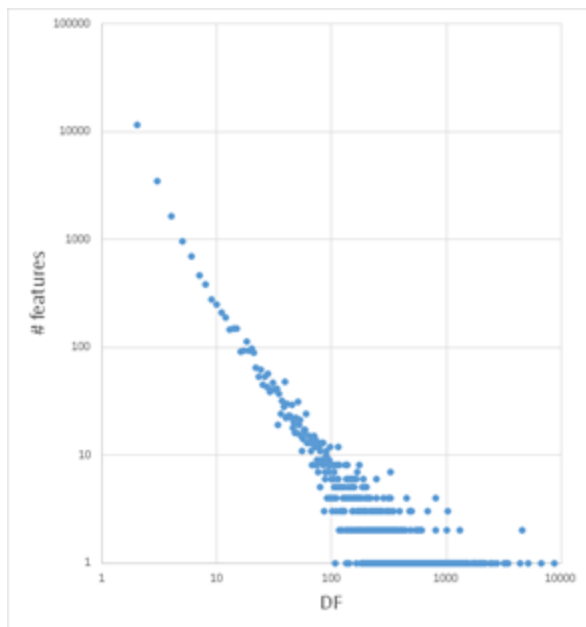
The use of a complex language is a **distinctive trait of humans**.

- Thousands of symbols.
- Complex syntax.
- Semantic is *mostly* compositional.
- Potentially ambiguous.
- Relies on shared world knowledge.
- Learned from scratch along life.
- It evolves along our lives.

James R Hurford, "Human uniqueness, learned symbols and recursive thought"
Thomas C. Scott-Phillips, Richard A. Blythe, "Why is combinatorial communication rare in the natural world, and why is language an exception to this trend?"

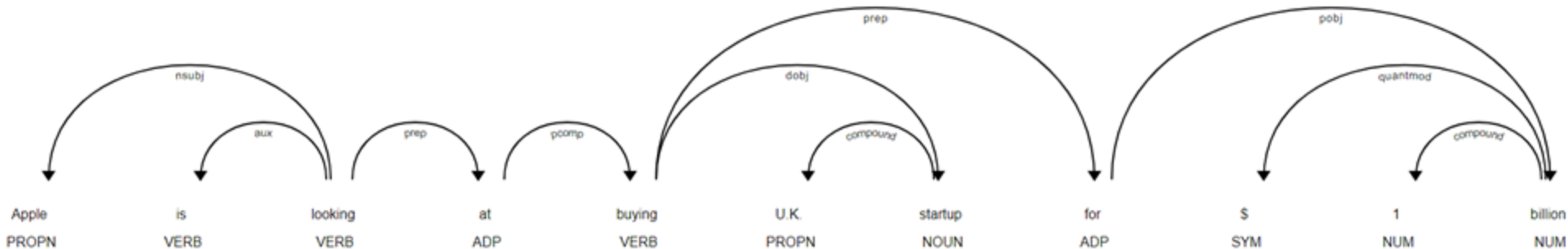
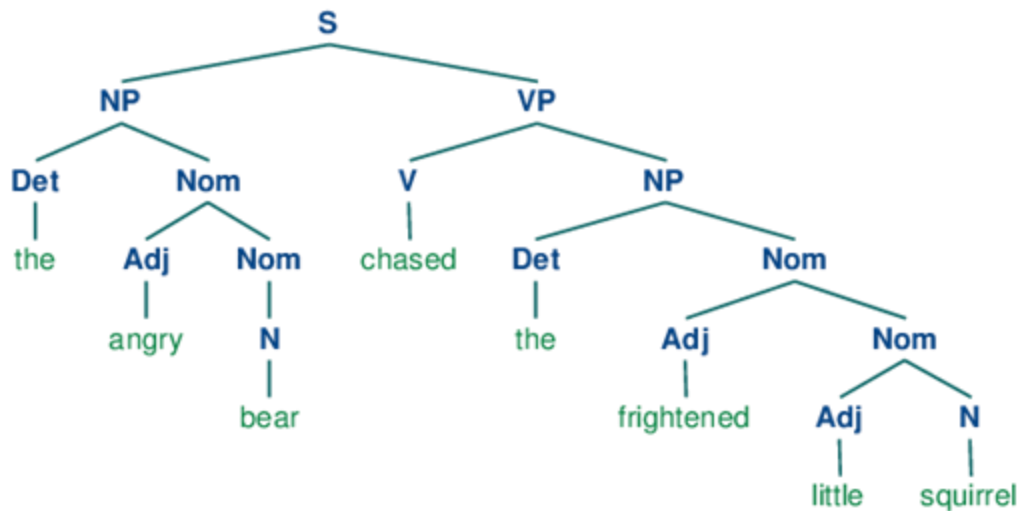
Natural Language

- Thousands of symbols.



Natural Language

- Complex syntax.



Natural Language

- Semantic is *mostly* compositional.

compositional:

to buy a car

leggere un libro

collocation:

make the bed

saltare la lezione

idiomatic expression:

break a leg

dalla padella nella brace

Natural Language

- Potentially ambiguous.
 - Part of speech ambiguity

beat: Verb, Noun?

Listen to this cool beat.

I'll beat you at checkers!

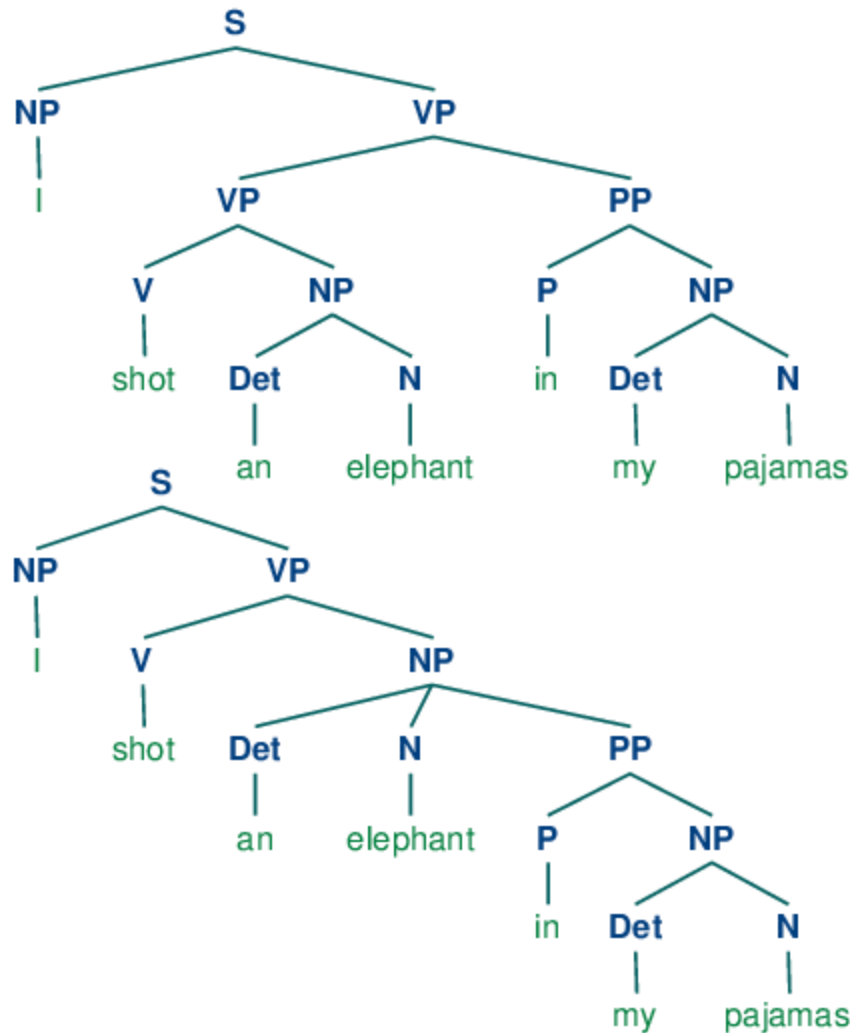
- Word sense ambiguity

interest: attention or money?

There is interest on this topic.

The bank raised the interest rate.

- Syntactic ambiguity →



Natural Language

- Relies on shared world knowledge.

"I'm faster than Lewis Hamilton"

Faster doing what?

"Alexis and Kate are sisters"

Relation.

"Alexis and Kate are mothers"

Same property, but no relation.

"My mother is younger than me."

Not possible.

"I was on a bike with a helmet"

The helmet is on my head.

"I was on a bike with an electric engine"

The engine is part of the bike.

Natural Language

- Learned from scratch along life.



Natural Language

- It evolves along our lives.

"airplane"

"quark"

"automobile"

"genocide"

"Tesla" (books, web)

"a ship"

"I googled it"

"it's in the cloud"

"crowdfunded"

"to navigate"

"super spreader"

"wear a mask"

Natural Language

The use of a complex language is a **distinctive trait of humans**.

- Thousands of symbols.
- Complex syntax.
- Semantic is *mostly* compositional.
- Potentially ambiguous.
- Relies on shared world knowledge.
- Learned from scratch along life.
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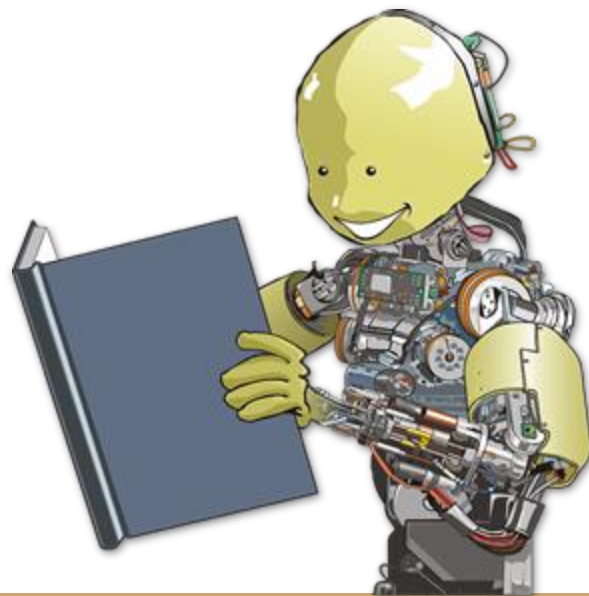
James R Hurford, "Human uniqueness, learned symbols and recursive thought"
Thomas C. Scott-Phillips, Richard A. Blythe, "Why is combinatorial communication rare in the natural world, and why is language an exception to this trend?"

Natural Language Understanding (by machines)

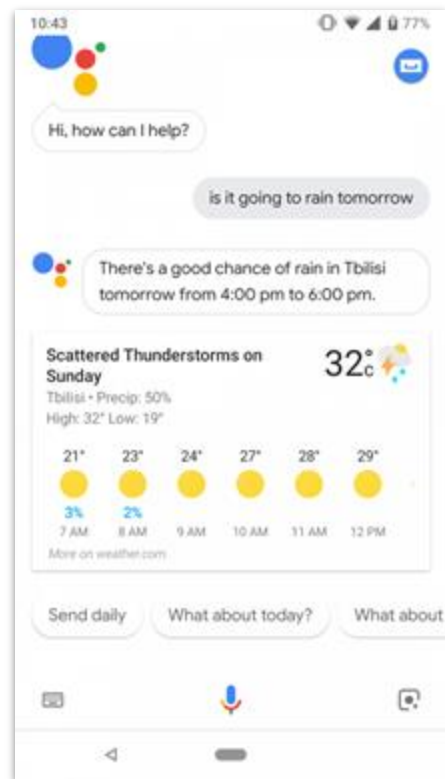
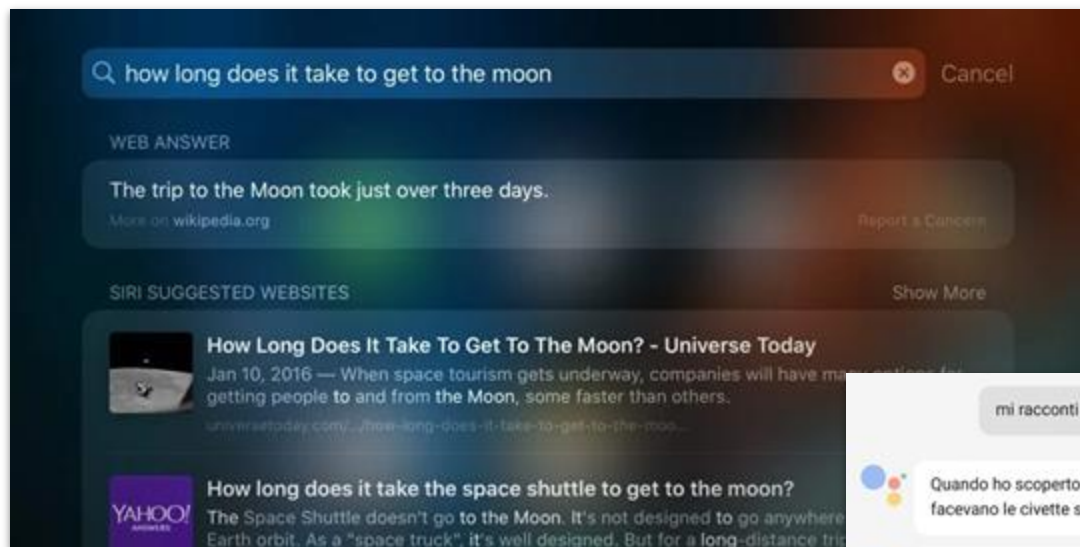
Natural Language Understanding aims at building machines that are able to receive and give information using natural language, like humans do.

From a computational point of view, natural language understanding is considered to be an AI-complete problem.

Practical text analytics tasks are simplifications of NLU that make the problem easier to solve.



Practical applications



Practical applications

volkswagen

All **News** Images Shopping Maps More ▾ Search tools

About 44,700,000 results (0.48 seconds)

 **Volkswagen** brakes as rivals race ahead in Europe
Financial Times - 7 hours ago
Volkswagen lost further market share in Europe in December as the rest of the world's manufacturers enjoyed a blockbuster end to the year that ...
Volkswagen's market share falls for first time since 2007
Irish Times - 3 hours ago
Volkswagen's European market share has slumped
Management Today - 2 hours ago
Only **Volkswagen** fails to cash in on rising European car sales
In-Depth - CTV News - 3 hours ago

CTV News Irish Times Management... 10News

[Explore in depth](#) (103 more articles)

 **Volkswagen's** Recall Proposal Isn't Good Enough for US ...
The Atlantic - Jan 13, 2016
By now, **Volkswagen's** explanations, apologies, lawsuits, and goodwill programs have all had their turn in the headlines. But a central question ...
Volkswagen's US recall proposals rejected
The Week UK - Jan 13, 2016
EPA Rejects **Volkswagen's** Proposed Diesel Emissions Fix
Carscoops (blog) - 15 hours ago
California board rejects **Volkswagen's** diesel fix plan
International - CNBC - Jan 12, 2016
Volkswagen, EPA Fail to Break Impasse in Emissions-Cheating ...
In-Depth - Wall Street Journal - Jan 13, 2016
Sweden prosecutor investigating **Volkswagen**
International - JURIST - Jan 12, 2016

Practical applications

🔍 Cerca su Twitter

Tendenze - Italia



1 · Di tendenza



#autocertificazione

1.615 Tweet

2 · Di tendenza



#Ratzinger

6.162 Tweet

3 · Di tendenza



#Pompeo

1.832 Tweet

4 · Di tendenza



#domenicain

5 · Di tendenza



#Hunziker

8.072 Tweet

[Mostra altro](#)

Tendenze - Spagna



1 · Di tendenza



Obtuve 12 de 13

1.864 Tweet

2 · Di tendenza



#EscapeRoomLDDSS

3 · Di tendenza



#VivaLaVida294

1.615 Tweet

4 · Di tendenza



#Iarlapardo82

5 · Di tendenza



Alvise

18.200 Tweet

[Mostra altro](#)

Tendenze - Giappone



1 · Di tendenza



#日向坂で会いましょう

26.300 Tweet

2 · Di tendenza



#乃木坂工事中

38.500 Tweet

3 · Di tendenza



#関ジャム

27.900 Tweet

4 · Di tendenza



#未来少年コナン

22.900 Tweet

5 · Di tendenza

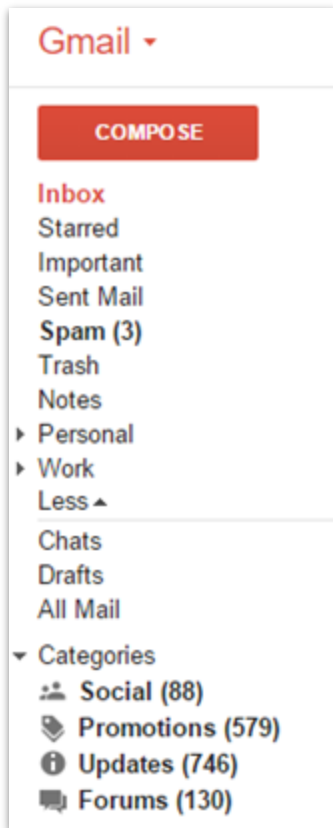


#京本大我入所14周年

13.900 Tweet

[Mostra altro](#)

Practical applications



Practical applications



The screenshot shows the Google Translate interface. At the top, there's a menu icon, the Google Translate logo, and a user profile icon. Below this, there are tabs for 'Text' and 'Documents'. The main area has a language selector with 'JAPANESE - DETECTED' on the left and 'ITALIAN' on the right. The Japanese text input is: 安倍晋三首相は、5月31日までの約1か月までに日本の緊急事態を延長し、新しいコロナウイルスと戦うことを計画していると政府当局者が日曜日に言った。 The Italian translation is: Il primo ministro Shinzo Abe ha in programma di estendere l'emergenza del Giappone di circa un mese entro il 31 maggio e combattere il nuovo coronavirus, ha detto domenica un funzionario del governo. There are also audio icons and a character count '72/5000'.

Google Translate

Text Documents

JAPANESE - DETECTED JAPANESE ITALIAN ENGLISH JAPANESE

安倍晋三首相は、5月31日までの約1か月までに日本の緊急事態を延長し、新しいコロナウイルスと戦うことを計画していると政府当局者が日曜日に言った。

Abe shinzō shushō wa, 5 tsuki 31-nichi made no yaku 1-kagetsu made ni Nihon no kinkyō jitai o enchō shi, atarashi koronaurusu to tatakau koto o keikaku shite iruto seifu

Show more

72/5000

Il primo ministro Shinzo Abe ha in programma di estendere l'emergenza del Giappone di circa un mese entro il 31 maggio e combattere il nuovo coronavirus, ha detto domenica un funzionario del governo.



The screenshot shows a tweet from Emmanuel Macron. The tweet text is in French: C'est un moment déterminant pour la communauté mondiale. En nous mobilisant aujourd'hui autour de la science et de la solidarité, nous semons les graines d'une plus grande unité demain. A Google Translate overlay is present, showing the English translation: This is a defining moment for the world community. By mobilizing ourselves today around science and solidarity, we will grow the seeds of greater unity tomorrow. Below the translation, there are links for 'Hide original' and 'Rate this translation'. At the bottom, there is a section titled 'Pour une mobilisation mondiale contre le virus.' followed by a quote in French and a 'See More' link.

Emmanuel Macron

9 hrs

C'est un moment déterminant pour la communauté mondiale. En nous mobilisant aujourd'hui autour de la science et de la solidarité, nous semons les graines d'une plus grande unité demain.

This is a defining moment for the world community. By mobilizing ourselves today around science and solidarity, we will grow the seeds of greater unity tomorrow.

Hide original Rate this translation

Pour une mobilisation mondiale contre le virus.

« La chance ne sourit qu'aux esprits bien préparés ». Ces mots sont ceux de Louis Pasteur, l'un des plus grands scientifiques au monde, à l'origine de découvertes et avancées majeures qui ont sauvé des millions de vies a...

See More

Practical applications



Relations with other data types: tabular data

Tabular data usually contain numerical, categorical, or symbolic values that have explicit and fixed structure, constraints and relations.

Extracting structured information from [historical text sources](#) may enable data mining on information not originally designed for such processes.

[Using natural language questions to query databases:](#)

Table

Rank	Name	No. of reigns	Combined days
1	Lou Thesz	3	3,749
2	Ric Flair	8	3,103
3	Harley Race	7	1,799
4	Dory Funk Jr.	1	1,563
5	Dan Severn	2	1,559
6	Gene Kiniski	1	1,131

Example questions

#	Question	Answer
1	Which wrestler had the most number of reigns?	Ric Flair
2	Average time as champion for top 2 wrestlers?	AVG(3749,3103)=3426
3	How many world champions are there with only one reign?	COUNT(Dory Funk Jr., Gene Kiniski)=2
4	What is the number of reigns for Harley Race?	7

Relations with other data types: graphs

Social networks are an important source of always up-to-date information about social groups, events and debated topics.

Methods based on **graph analysis** enables discovering the **structure and the dynamics** of a network at various levels: local relations, communities, large scale behaviors.

Enriching these methods with the **analysis of the actual content** that is shared on a network enables to better characterize the elements of the network and to give more detailed meaning to the interactions, enabling to investigate complex processes, e.g., abnormal behaviors, manipulation attempts...



Relations with other data types: images and video

Cross-media processing:

- Image captioning is the task of describing the content of an image.

It has application in retrieval processes.

It is of help for people with vision problems.



Describes without errors

Describes with minor errors

Somewhat related to the image

Unrelated to the image

Relations with other data types: images and video

Cross-media processing:

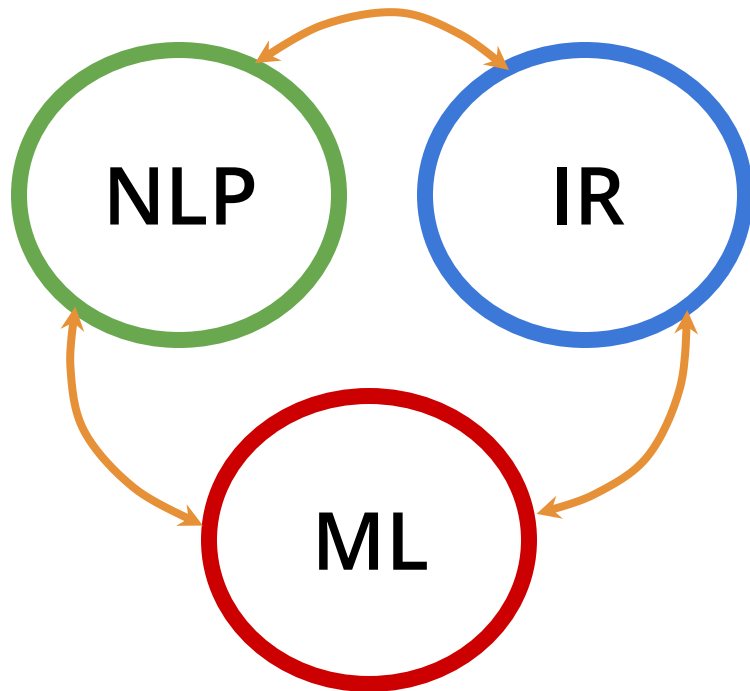
- Abstract visual representation of text enables smarter retrieval methods.



Technologies

These disciplines are not one distinct from the other.

- NLP methods are often built themselves on top of IR and ML methods.
- ML adopts IR measures to define the goals of the learned models.
- ML often assumes language to be manipulated by NLP and IR methods to be able to work on it.



Pipelines

The concept of a **processing pipeline** is recurring in text analytics methods, as text is rarely processable *as is* by ML methods and it requires to be transformed into representation that are fit to the specific ML method in use.

Such transformations are not just simple conversions of format, they may actively filter and modify the information in order to be better exploited by successive stages of the pipeline, so as to improve the final outcome of the process.



Text from the Web

Getting text from the Web

The [urllib](#) package implements methods that enable low-level interaction with Web servers.

Getting any Web page content is typically simple:

```
from urllib import request
url = "http://hpc.isti.cnr.it/~nardini/"
response = request.urlopen(url)
html = response.read().decode('utf8')
html
```

```
Out: '<!DOCTYPE html>\n<html lang="en-US">\n<head>\n<meta
charset="UTF-8" />\n<title>Franco Maria Nardini</title>\n<link ...'
```

Extracting (clean) text from the Web

The [BeautifulSoup](#) package implements methods to navigate, modify, and extract data from HTML and XML data.

It can be used on Web pages to extract information of interest.

```
from urllib import request
url = "http://hpc.isti.cnr.it/~nardini/"
response = request.urlopen(url)
html = response.read().decode('utf8')
```

```
from bs4 import BeautifulSoup
soup = BeautifulSoup(html, "html5lib")
```

```
soup.get_text()
```

← *simple way or ↓ more control*

```
[ ''.join(s.findAll(text=True)) for s in soup.findAll('p') ]
```

NLTK

NLTK

[NLTK](#) is an open source library that support quick development of NLP applications.

[NLTK comes with an online book](#) that has guided examples for many tasks.

Typical installation steps:

```
>conda install nltk
```

```
>python
```

```
In : import nltk
```

```
In : nltk.download()
```

(pick option to download popular packages)

Tokens & Sentences

Tokenization

One of the basic operation on text is to identify the words (tokens) composing it.

NLTK provides a basic function to perform a *language aware* tokenization.

```
> from nltk import word_tokenize  
> tokens = word_tokenize(text)  
> tokens[1000:1050]
```

```
Out: ['with', 'you', 'and', 'you', 'are', 'no', 'longer', 'my',  
'friend', ',', 'no', 'longer', 'my', "'", 'faithful', 'slave',  
,',', "'", 'as', 'you', 'call', 'yourself',... ]
```

Sentence splitting

Splitting a text into its sentences is not always a trivial task.

Use of punctuation is the most relevant clue, yet its use to report other information (acronyms, numbers, initials...) can produce erroneous splits.

I'll take the 8.30 train to St. Louis... or maybe I'll stay in LA. Time will tell!

Which of the above dots separate sentences?

The `sent_tokenize` function implements a language-aware sentence splitter.

Tokenization

The [NLTK tokenize](#) package implements tokenizers and splitters for a number of different cases:

- twitter text (very informal text, and platform specific style)
- multi-word expressions (providing a list of MWE in input)
- based on regular expressions
- based on [Stanford CoreNLP](#) tool
- other dataset-specific tokenizers

Some basic explorations

Given a tokenized text, we may want to extract some information out of it:

- build a vocabulary of the terms used in it
- count frequency of use of each term
- plot the distribution of frequencies across terms.
- see where terms appear across the text
- see in which contexts terms are used

The [Text](#) and [FreqDist](#) objects from NLTK are two simple tools to perform this tasks.

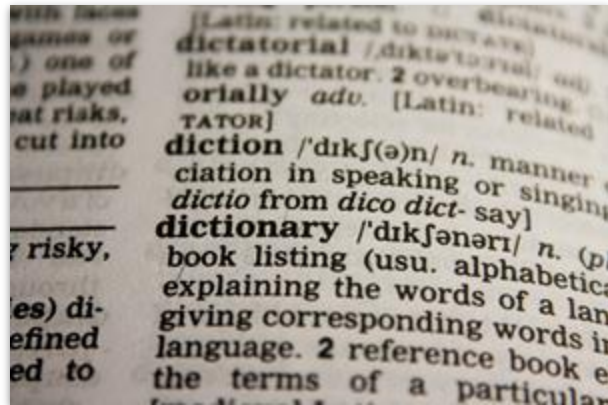
Stems & Lemmas

Stemming and lemmatization

Stemming and lemmatization aim at reducing the different inflections of a words to its root.

Stemming does it by applying a set of language-dependent word transformation rules, which can result in a lexically incorrect or non-meaningful word.

Lemmatization use a deeper, and costlier, NLP analysis to reduce a word to its dictionary form.



Stemming and lemmatization

Stemming:

```
In : from nltk.stem.porter import PorterStemmer
```

```
In : stemmer = PorterStemmer()
```

```
In : stemmer.stem('cars')
```

```
Out: 'car'
```

```
In : stemmer.stem('was')
```

```
Out: 'wa'
```

Stemming and lemmatization

Lemmatization:

```
In : from nltk.stem.wordnet import WordNetLemmatizer
```

```
In : lmtzr = WordNetLemmatizer()
```

```
In : lmtzr.lemmatize('cars')
```

```
Out: 'car'
```

```
In : lmtzr.lemmatize('was') # needs POS
```

```
Out: 'wa'
```

```
In : lmtzr.lemmatize('was', pos='v')
```

```
Out: 'be'
```

Bag of Words

From list of words...

Features are extracted from text in a list, possibly with repetitions.

```
In : text = 'the president of the united states of america'
```

```
In : feats = word_tokenize(text)
```

```
In : len( feats )
```

```
Out: 8
```

```
In : feats
```

```
Out: ['the', 'president', 'of', 'the', 'united', 'states', 'of',  
      'america']
```

Representing text with that **list of variable length** can be a problem for many ML algorithms.

...to Bag of Words

The most frequently adopted model is the Bag of Words (BOW) model, i.e., a document is represented by the set of words it is composed of.

```
In : bow = set(feats)
```

```
In : len(bow)
```

```
Out: 6
```

```
In : bow
```

```
Out: ['america', 'of', 'president', 'states', 'the', 'united']
```


...to Bag of Words

BOW loses information on word frequency.

Counts of word occurrences in every document can be stored in additional data structures.

Similarly to unigram models, BOW loses word order information.

N-grams can be used also in this case.

BOW representation has a fixed length, easier to manage by ML algorithms.

The set of all the distinct extracted features may be called 'vocabulary', 'dictionary', 'feature set', 'feature space'.

Bag of Words

BOW loses the frequency and word order information.

```
In : t1 = set(word_tokenize('I won, and thus you lose.'))
```

```
In : t2 = set(word_tokenize('I lose, and thus you won.'))
```

```
In : t1
```

```
Out: ['and', 'lose', 'thus', 'won', 'you']
```

```
In : t2
```

```
Out: ['and', 'lose', 'thus', 'won', 'you']
```

```
In : t1==t2
```

```
Out: True
```

N-grams

Word n-grams

Information on positional relations can be extracted with *Word n-grams* features, which capture local word order.

```
In : t1 = set(nltk.ngrams(nltk.word_tokenize('I won, and thus, you lose.'),2))
In : t2 = set(nltk.ngrams(nltk.word_tokenize('I lose, and thus, you won.'),2))
In : t1, t2, t1==t2
```

```
Out: {'W2G_and_thus', 'W2G_thus_you', 'W2G_won_and', 'W2G_you_lose', 'and', 'lose',
      'thus', 'won', 'you'},
```

```
Out: {'W2G_and_thus', 'W2G_lose_and', 'W2G_thus_you', 'W2G_you_won', 'and', 'lose',
      'thus', 'won', 'you'}
```

```
Out: False
```

Feature format ('W2G_...') doesn't matter as long as there is no ambiguity between features extracted by different methods.

Character n-grams

Character n-grams mitigate the effect of typos.

```
In : t1 = set(nltk.ngrams('rainbow',3))
```

```
In : t2 = set(nltk.ngrams('rainbaw',3))
```

```
In : t1, t2
```

```
Out:{'C3G_a_i_n', 'C3G_b_o_w', 'C3G_i_n_b', 'C3G_n_b_o',  
      'C3G_r_a_i', 'rainbow'}
```

```
Out:{'C3G_a_i_n', 'C3G_b_a_w', 'C3G_i_n_b', 'C3G_n_b_a',  
      'C3G_r_a_i', 'rainbaw'}
```

```
In : t1.intersection(t2)
```

```
Out: {'C3G_a_i_n', 'C3G_i_n_b', 'C3G_r_a_i'}
```

Zipf law

The distribution of words in text follows a [Zipf law](#).

313 words generate half of the 500k occurrences in Reuters 21578.

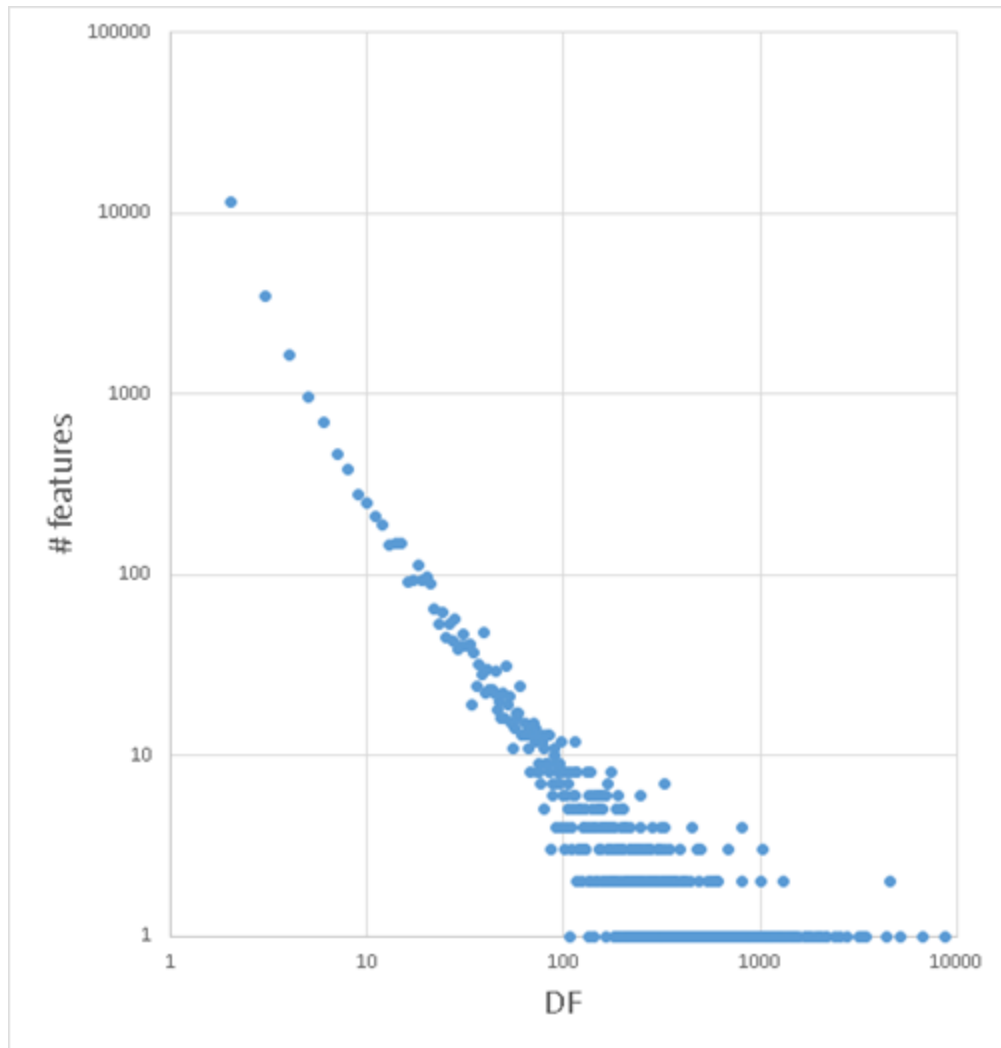
Frequency of use is inversely proportional to rank by frequency: $f \propto 1/r$

Principle of least effort: both the speaker and the hearer in communication try to minimize effort:

- Speakers tend to use a small vocabulary of common (shorter) words
- Hearers prefer a large vocabulary of rarer less ambiguous words
- Zipf's law is the result of this compromise

Related laws:

- Number of meanings m of a word obeys the law: $m \propto 1/f$
- Inverse relationship between frequency and length



Stopwords

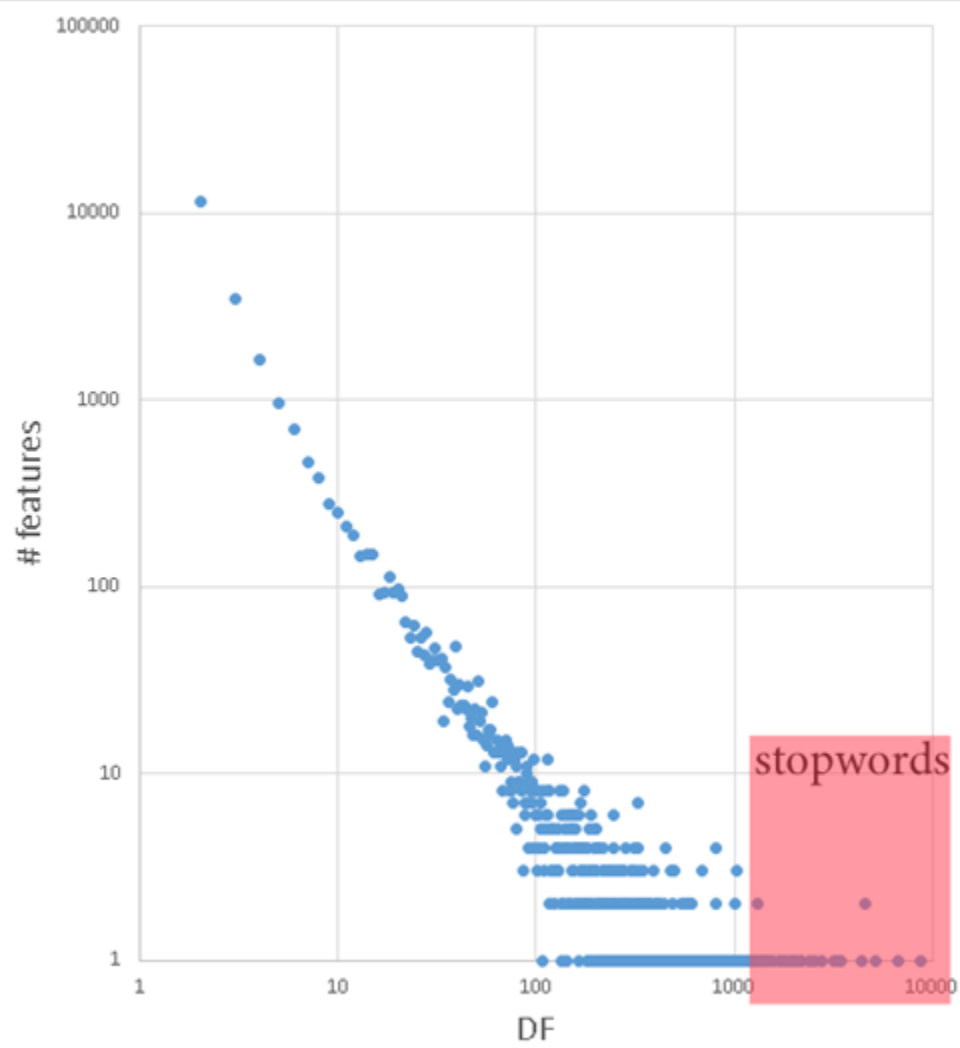
How much words like *of, the, a, an, to*, contribute to the semantic of a document?

the president of the united states of america
president united states america

The most common words of a language are usually referred as *stopwords*, i.e., words that can be removed from text without losing much information.

Stopwords lists are commonly provided by many NLP tools, yet they may not fit all the applications.

For example, MySQL's stopword list includes words like *appreciate, serious* and *unfortunately*, which are relevant for sentiment classification.



Stopwords

```
In : import nltk
```

```
In : from nltk.corpus import stopwords
```

```
In : stopwords.words('english')
```

```
Out: ['i', 'me', 'my', 'myself', 'we', 'our', 'ours', ' ourselves',  
...]
```

```
In : features = set(nltk.word_tokenize('the president of the united  
states of america'))
```

```
In : less_features = features.difference(stopwords.words('english'))
```

```
In : less_features
```

```
Out: {'america', 'president', 'states', 'united'}
```

Rare features

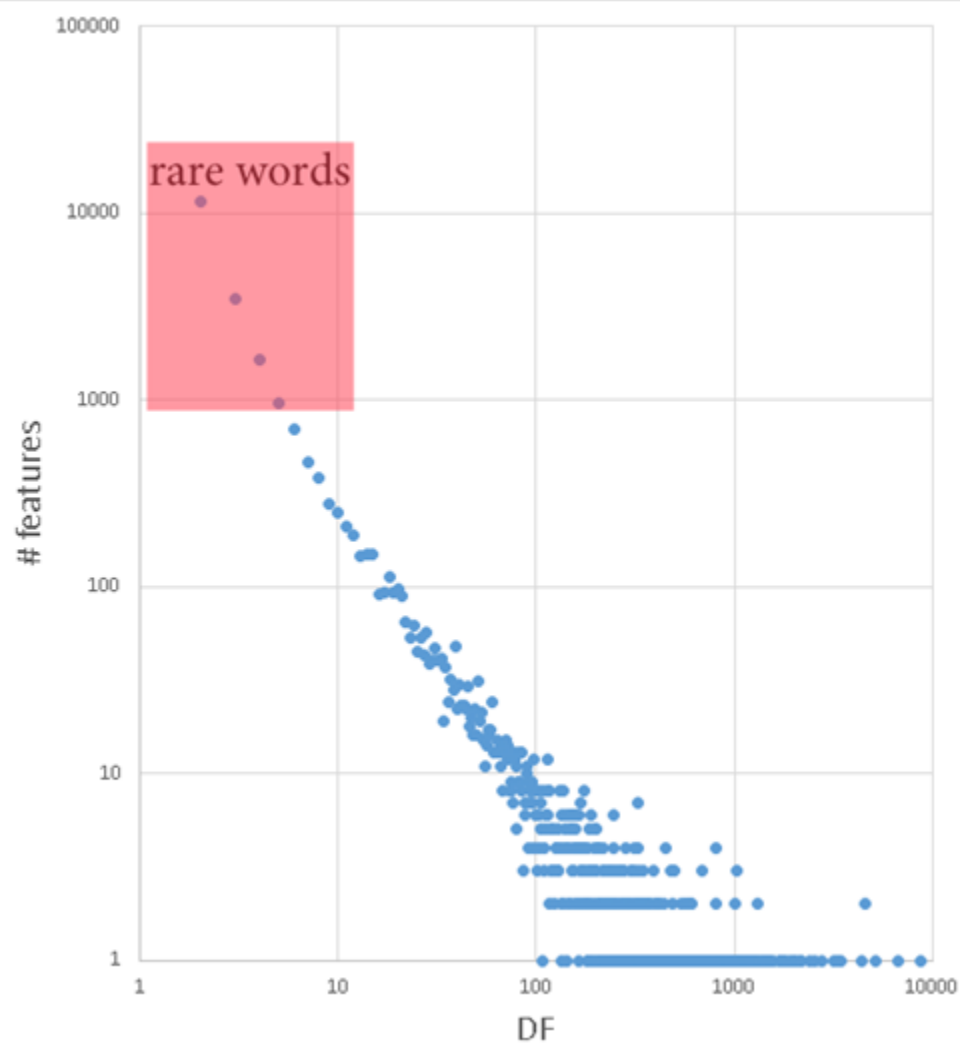
Features that appear in very few documents do not bring useful information to generalize the TM problem we are facing.

If a word appears rarely in text observed in the past, it will likely be rare in future text. Making it of little help to process most of the future input text.

A word can be rare because it is a random typo, or some kind of artificially built identifier that is bound to the document, e.g., a [slug](#).

Words that appear only once usually account for a large portion of the distinct words appearing in a text collection.

Removing rare words makes it faster to process the indexed text, also requiring less memory space.



Vector Space Model

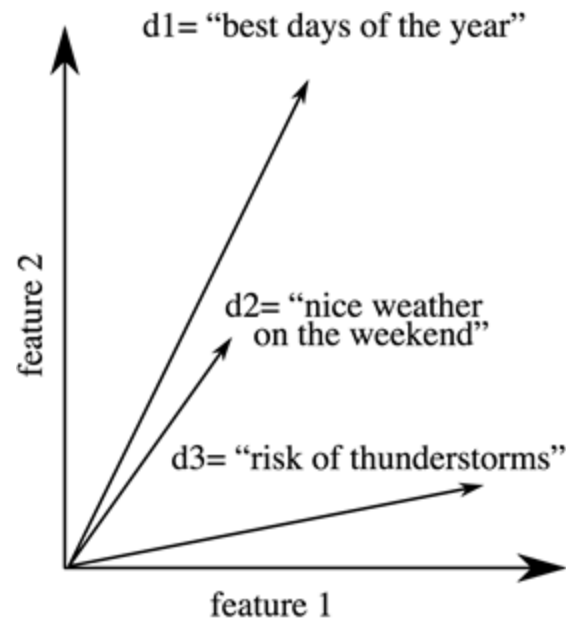
Vector Space Model

We now have each document represented as a $|F|$ -dimensional vector w , where $|F|$ is the number of distinct features extracted from text.

Documents whose vector are close one to each other are likely to be similar.

w_{df} indicates the relevance of the feature f in the document d .

Vectors are usually **sparse**, i.e., most of their values are zero.



Vector Space Model

Each feature is mapped to a distinct dimension in $R^{/F/}$ using a *one-hot* vector:

$$v('I') = [1, 0, 0, \dots, 0, 0, \dots, 0, 0, 0]$$

$$v('you') = [0, 1, 0, \dots, 0, 0, \dots, 0, 0, 0]$$

$$v('won') = [0, 0, 1, \dots, 0, 0, \dots, 0, 0, 0]$$

$$\vdots$$

$$v('a_i_n') = [0, 0, 0, \dots, 0, 1, \dots, 0, 0, 0]$$

$$\vdots$$

$$v('NN') = [0, 0, 0, \dots, 0, 0, \dots, 0, 0, 1]$$

Vector Space Model

A document is represented as the weighted sum of its features vectors:

$$v(d) = \sum_{f \in d} w_{fd} v(f)$$

For example:

$$\begin{aligned} d &= \text{'you played a good game'} \\ v(d) &= [0, w_{\text{played},d}, w_{\text{game},d}, 0, \dots \dots 0, w_{\text{good},d}, 0 \dots \dots 0, 0] \end{aligned}$$

The resulting document vectors are sparse:

$$|\{i | v_i(d) \neq 0\}| \ll n$$

How do set the weights w_{fd} ?